An Introduction to 3D Printing

Part 2: Practical Application

Examples, Critical Thinking, and Design

Outline

Part 1: Printers
Process
Limitations
Dimensional Accuracy
Advantages

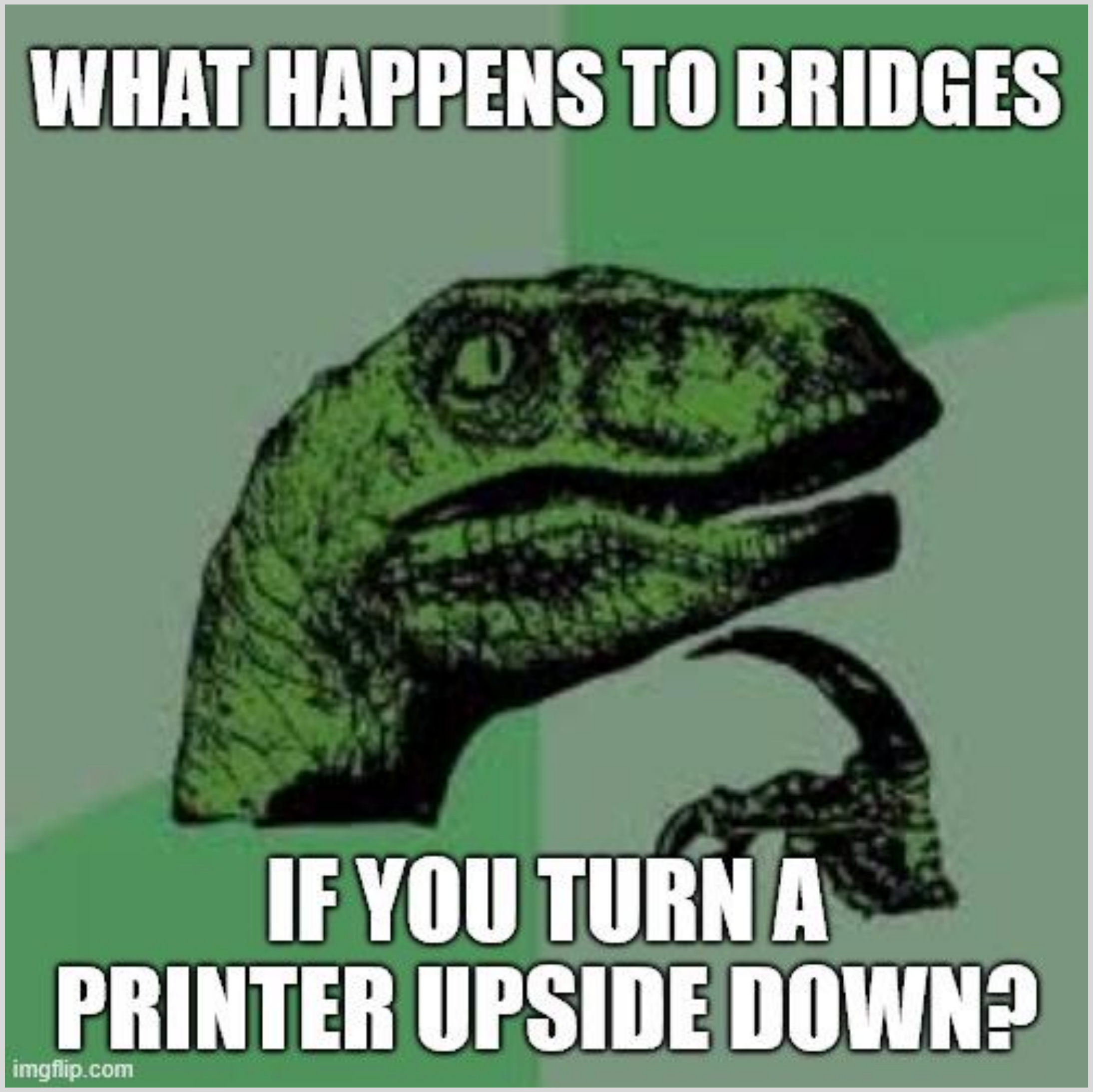
Part 2: Practical Application
Examples
Critical Thinking
Design

Part 3: Science
Vibrations and Waves
Linear Motion
Thermal Expansion

Pop Quiz

- How can you help printed parts fit together?
  - You sometimes need scale factor.
  - You always need constant offset. Start with 0.25 mm for XY, one layer for Z.

Anti-Gravity

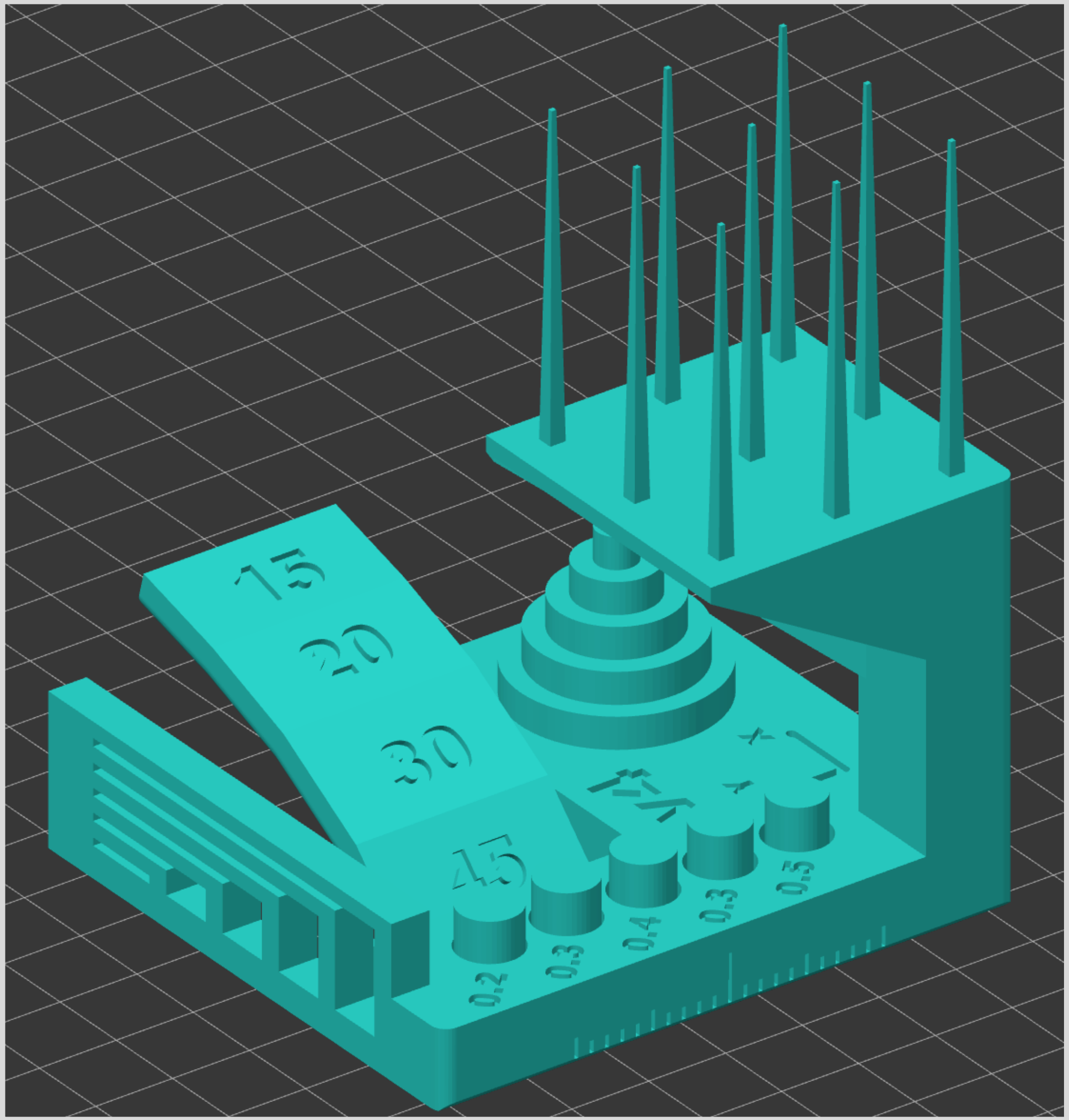


Anti-Gravity

- Positron Github.
  - Prints upside down.
  - Folds up and fits in a small box.
- Design team says same print quality. Why?
- Extrusions hold their shape when you press them onto a solid surface.

Critical Thinking: Stress Testing

- Autodesk Test Print. One of many test prints.
- Prints that highlight **capabilities** and **limitations** of your printer + filament.
- What are these shapes testing?
- Full breakdown here.
- Bridging, overhangs, PIP tolerance, dimensional accuracy, …



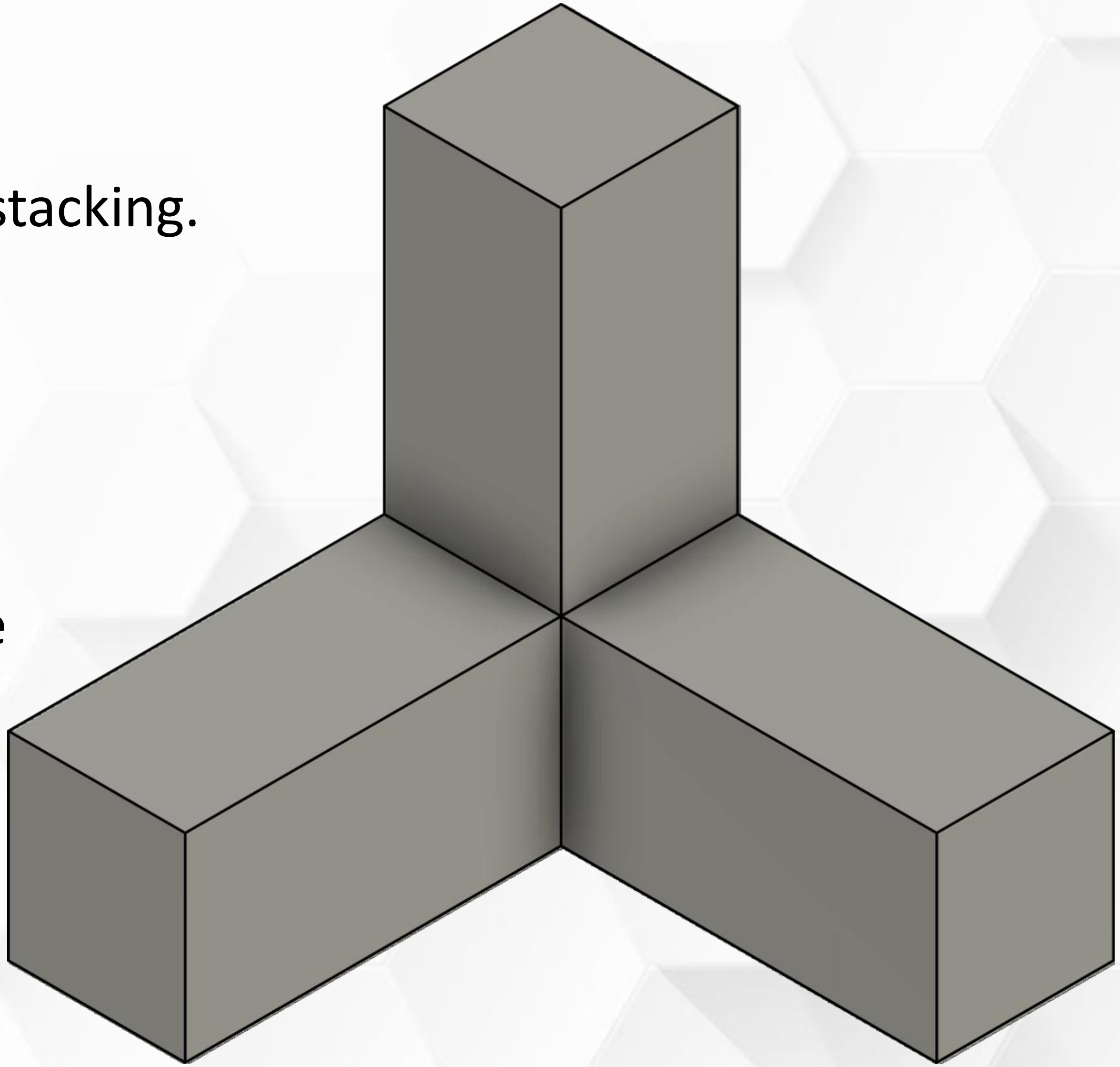
Critical Thinking: Corners

- Rounding corners with fillets is a good idea. Why?
- Stresses concentrate in corners.
  - Adding fillets improves strength.
  - Fillets on bottom surfaces reduce first layer warping.
- Print faster with more accurate movements. Why?
  - Think of a racecar (toolhead) going around a track.
  - Smoothing changes of direction reduces vibrations.

Critical Thinking: Corner Bracket

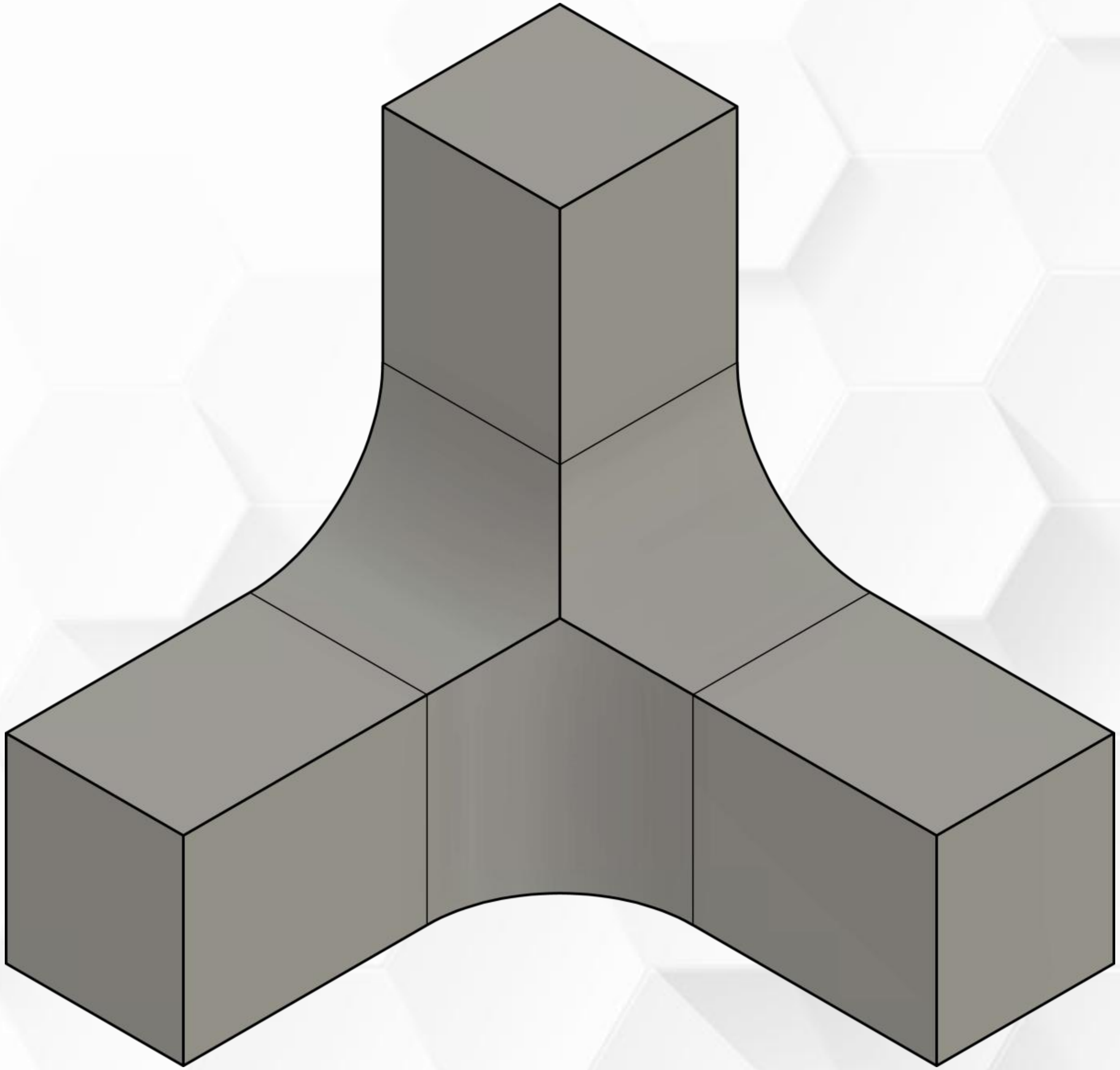
Consider this corner bracket.

- Top section is weaker because of layer stacking.
- How can you improve strength?
    - Stronger filament.
    - Stronger slicer settings.
    - What else?
- Let’s rethink the design and make some modifications.



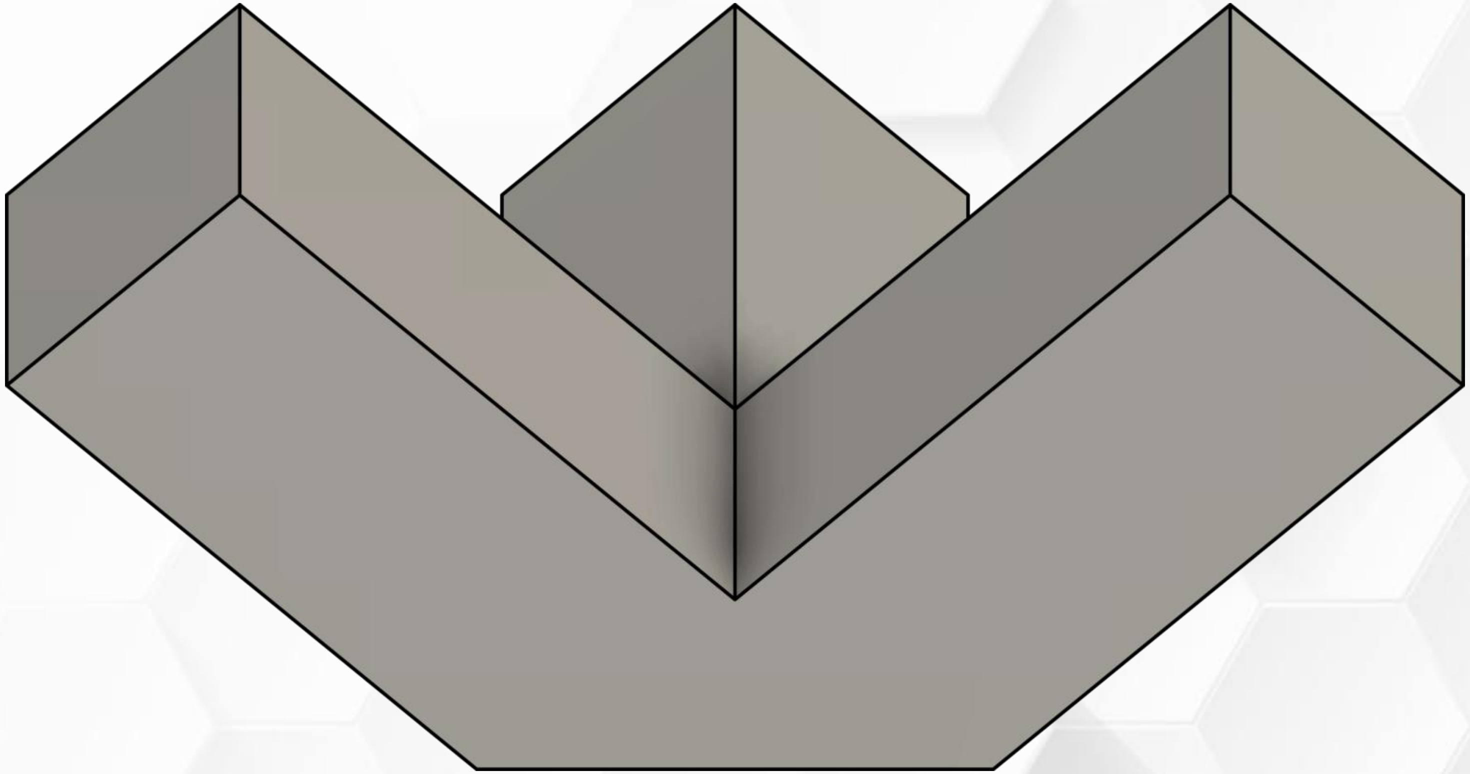
Critical Thinking: Corner Bracket

- Reinforce joints with **fillets**.
- Reduced stress concentration.
- Reinforced the weakest point.



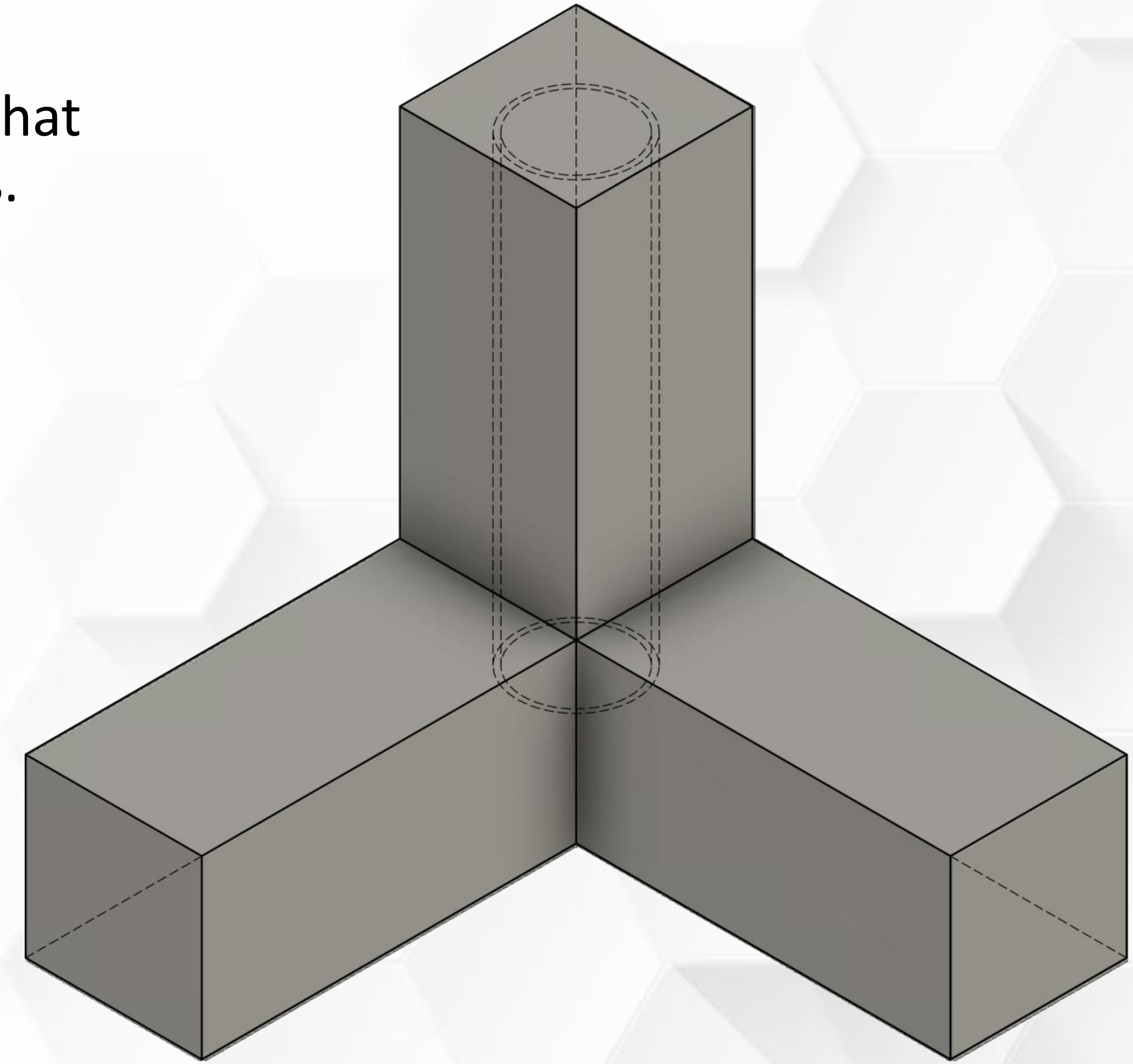
Critical Thinking: Corner Bracket

- Trim the corner to create an angled print orientation.
- Add overhangs.
- Reduced first layer contact.
- Each section has equal strength.
- Slightly improved layer bonding.



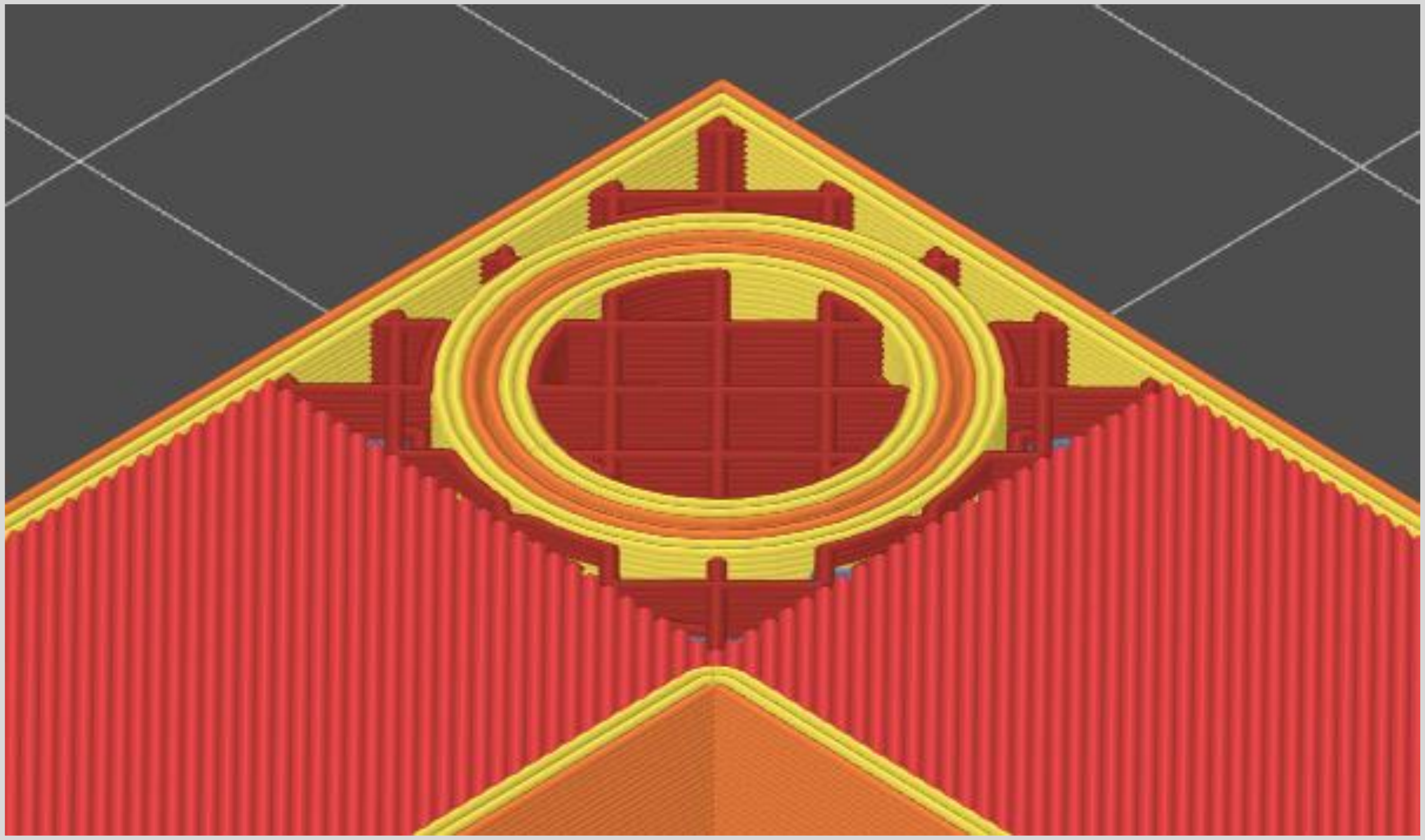
Critical Thinking: Corner Bracket

- Cut a small slit (around 0.1 mm wide) that connects the upper and lower sections.
- Why?
- *Hint:* What will the slicer do?



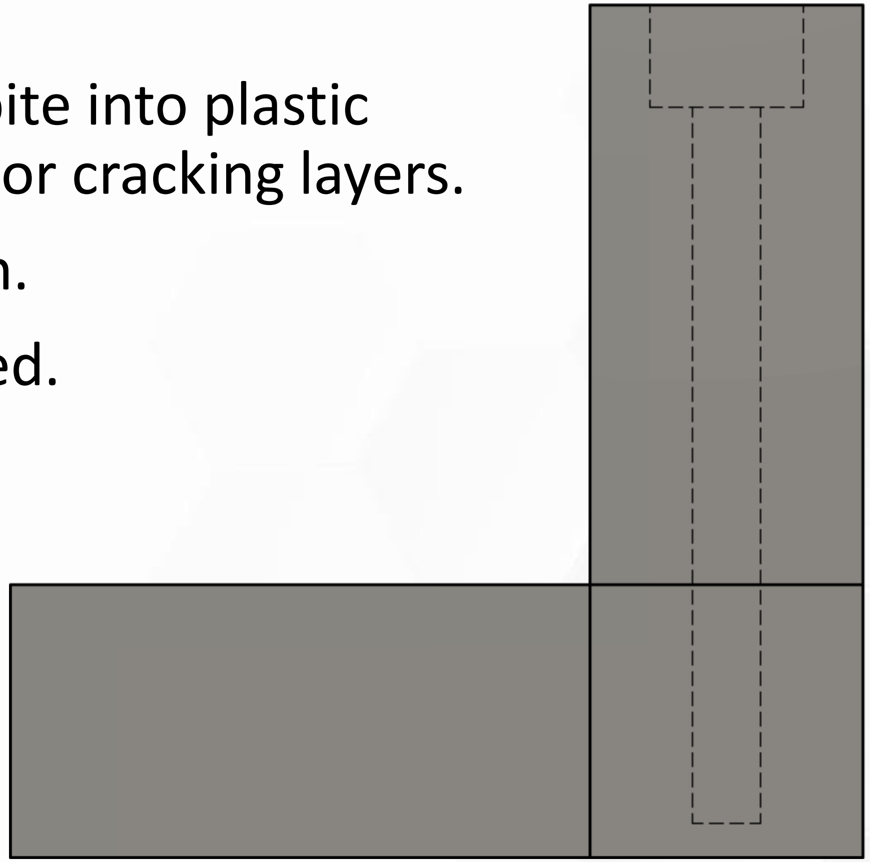
Critical Thinking: Corner Bracket

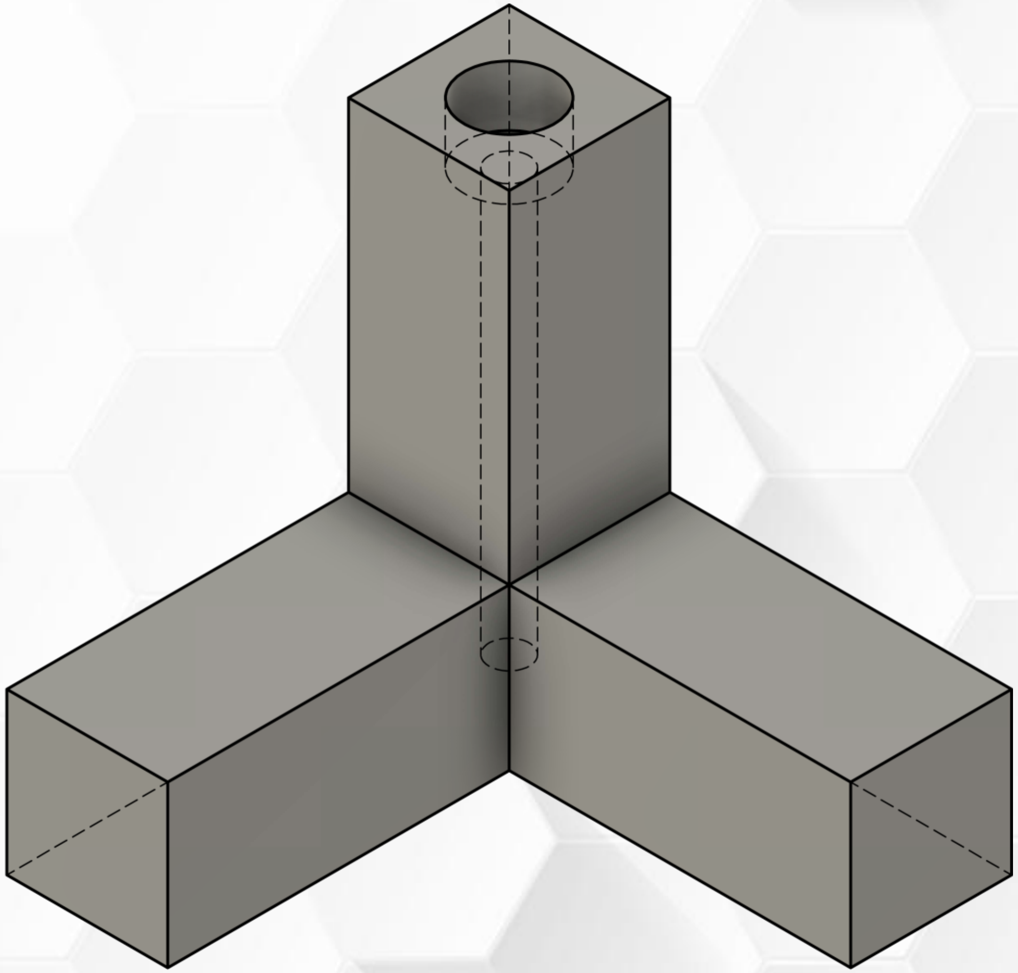
- Tricks slicer into adding extra perimeters.
- Increased layer bonding.
- Looks the same from the outside.



Critical Thinking: Corner Bracket

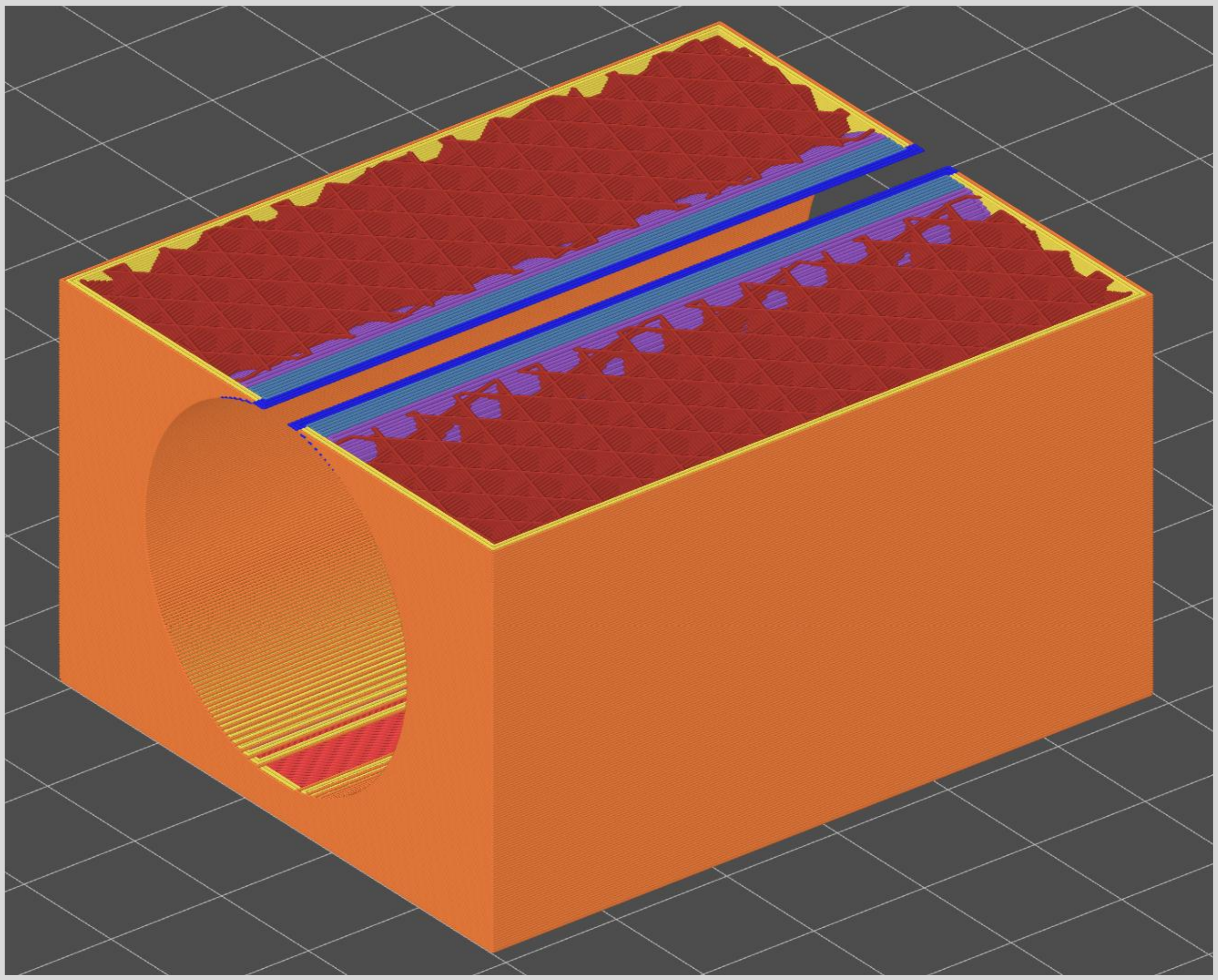
- Cut a hole. Add tight fitting screw after printing.
- Threads should bite into plastic without splitting or cracking layers.
- Superior strength.
- Assembly required.





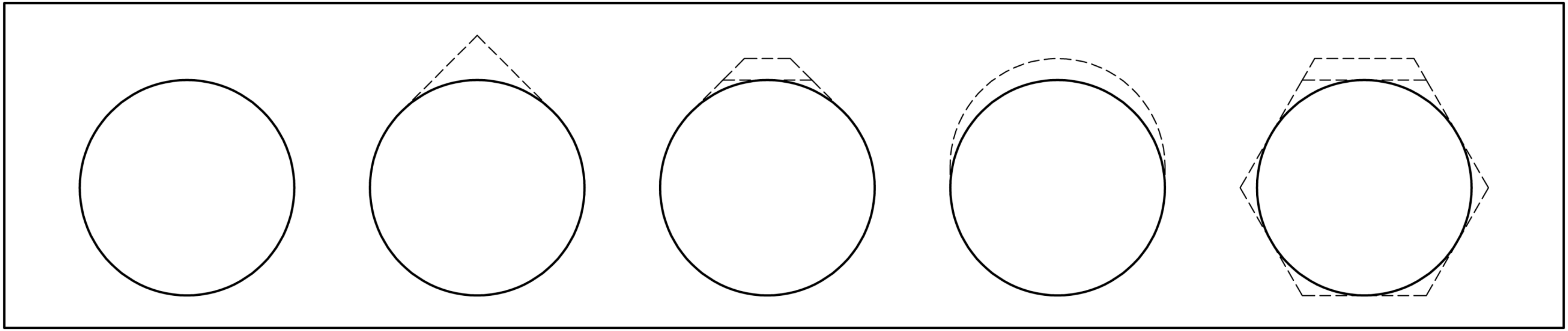
Critical Thinking: Horizontal Holes

- Horizontal holes have steep overhangs at the top.
- If extrusions collapse, the hole is undersized.
- Assume the hole must remain horizontal.
- How can you alter the design to fix the problem?



Critical Thinking: Horizontal Holes

- Make the top profile more printer-friendly.
- Add a clearance at the top.



Critical Thinking: Bridges

- What can you do to improve bridges, overhangs, and floating extrusions? (Besides supports.)
- *Hint:* How can you help extrusions cool and lock in place.
    - Lower filament temperature.
    - Slower printing speed.
    - Increase cooling.
- Reimagine the object as multiple pieces.
    - Each piece is free from supports and bridges.
    - Assembly and clearances required.

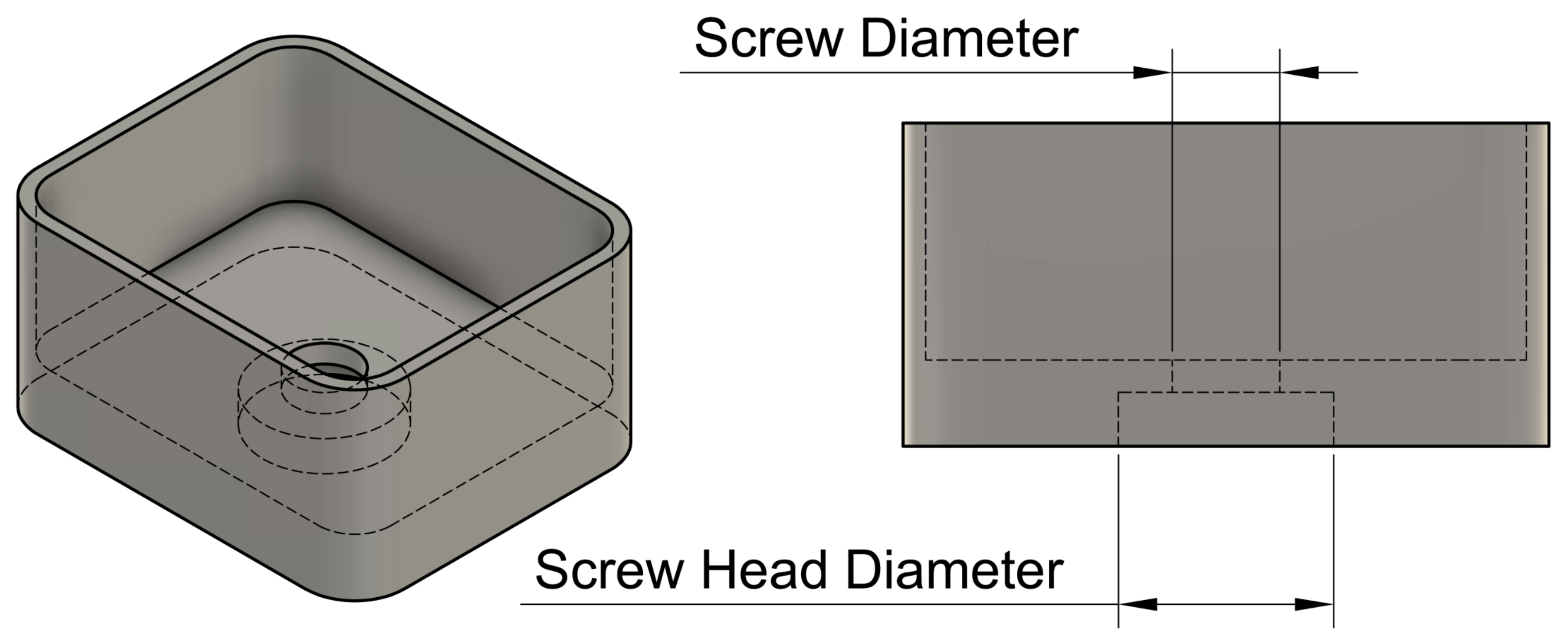
Sequential Bridges

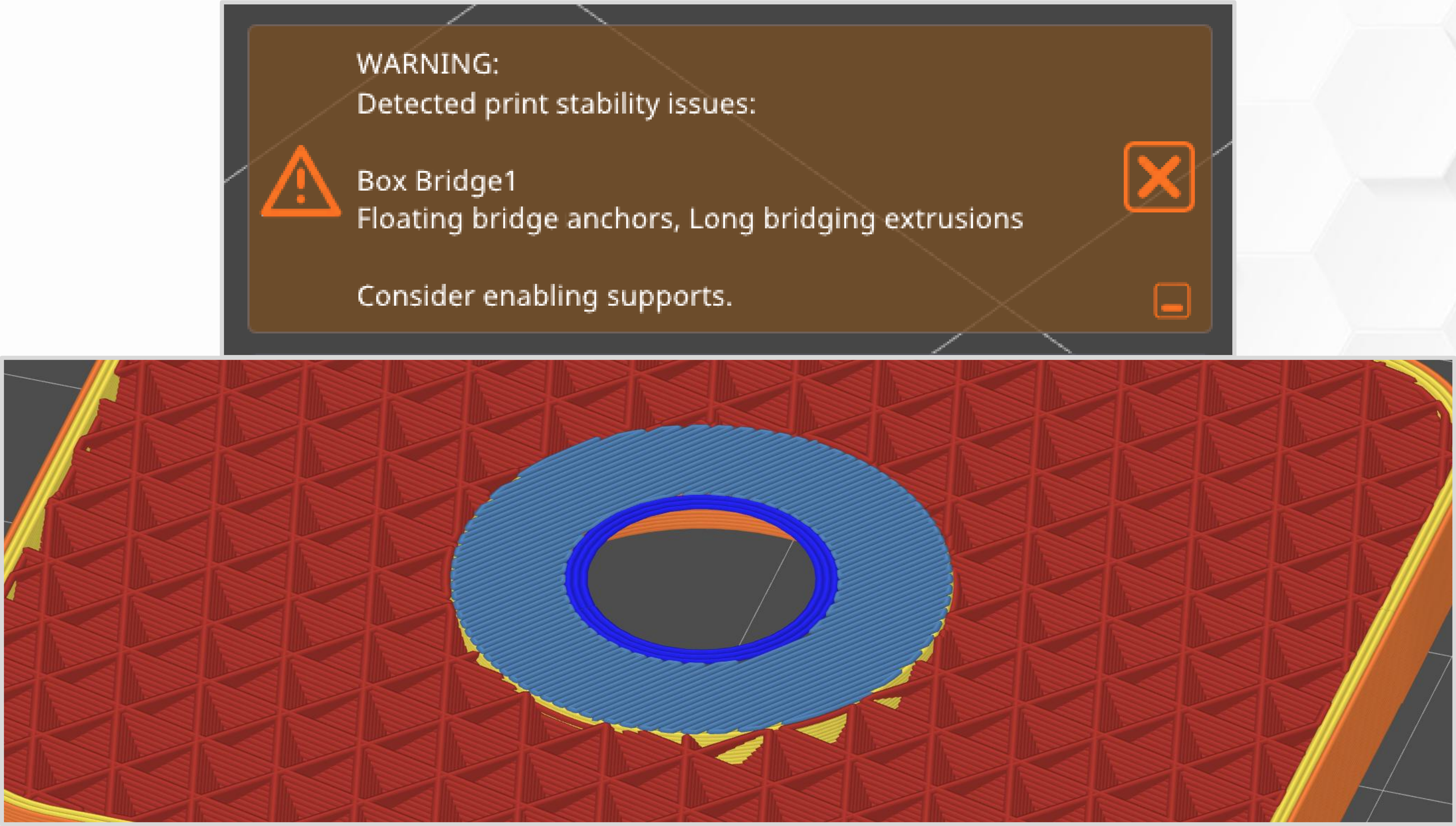
Let’s combine small bridges to create a larger, multi-step bridge.

Critical Thinking: Sequential Bridges

Consider a storage box with a countersunk screw on the bottom.

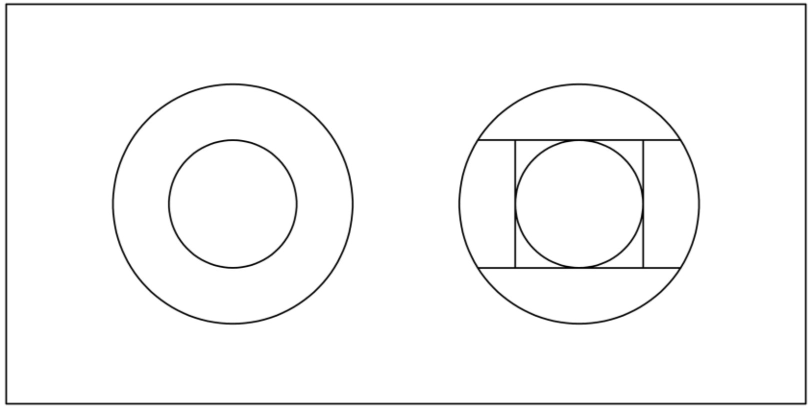
- Can you see the hidden problem?

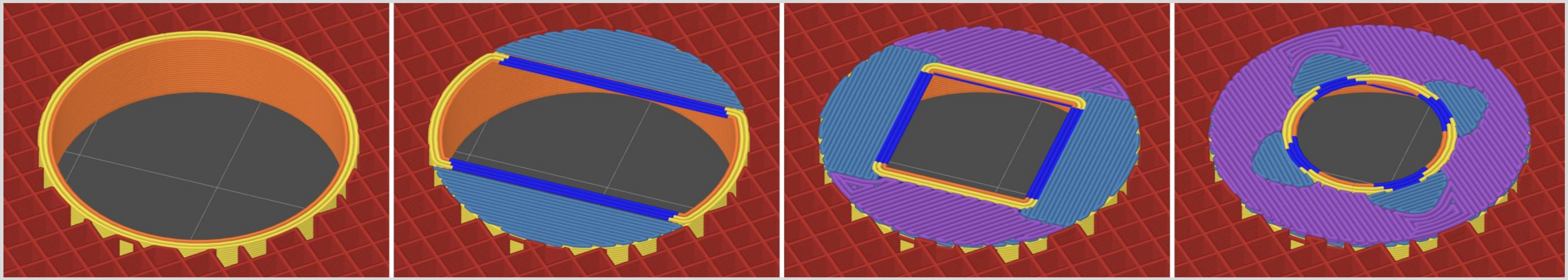



WARNING:
Detected print stability issues:
Box Bridge1
Floating bridge anchors, Long bridging extrusions
Consider enabling supports.

Critical Thinking: Sequential Bridges

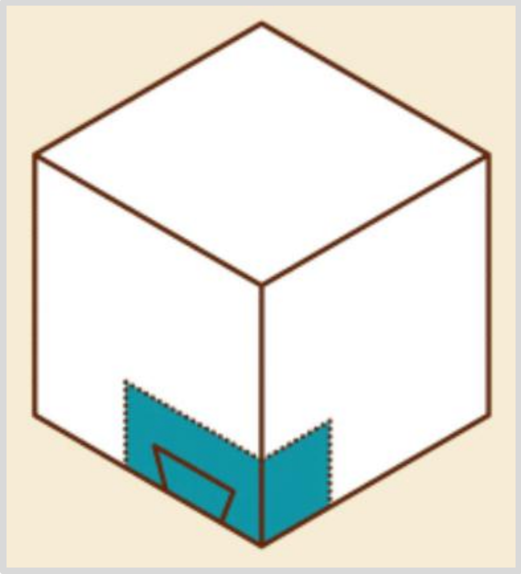
- There are floating extrusions!
- We can fix this without supports.
- Complete the bridge in multiple steps.

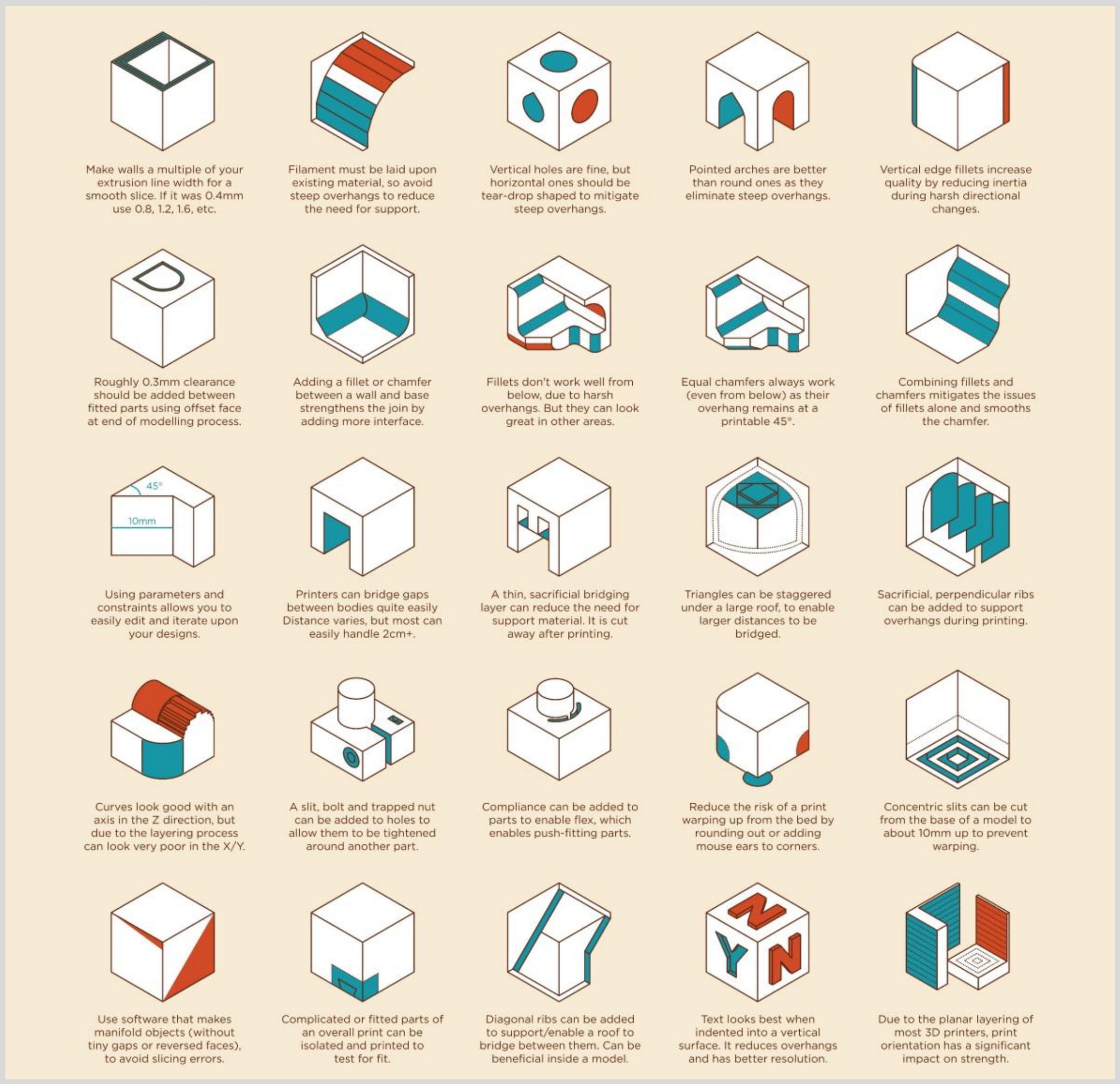




More Design Tips

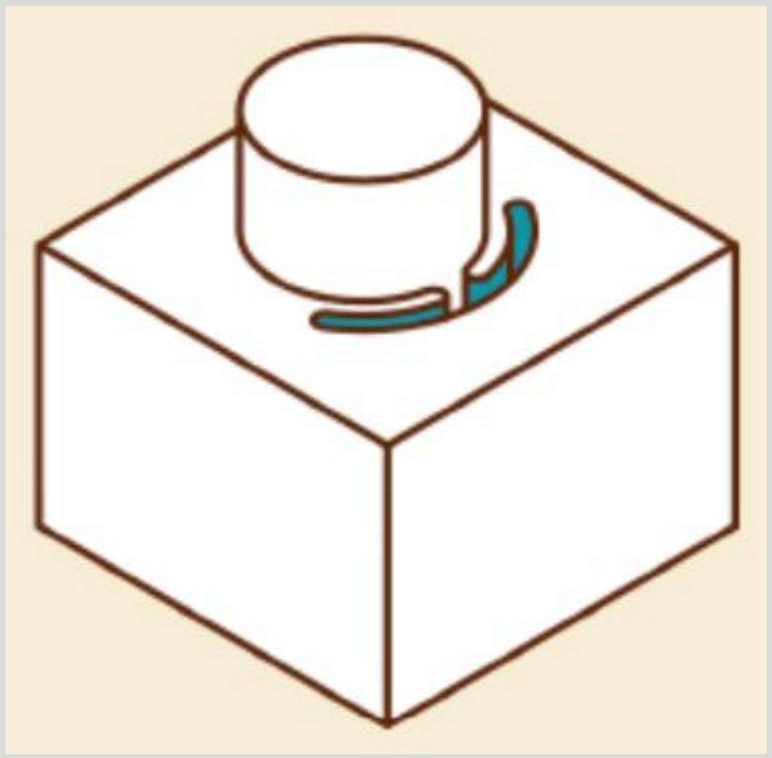
- CAD Design Tips for 3D Printing by Billie Ruben.
- “Complicated or fitted parts of a print can be isolated and printed to test for fit.”





Even More Design Tips

- “Compliance can be added to parts to enable flex, which enables push-fitting parts.”



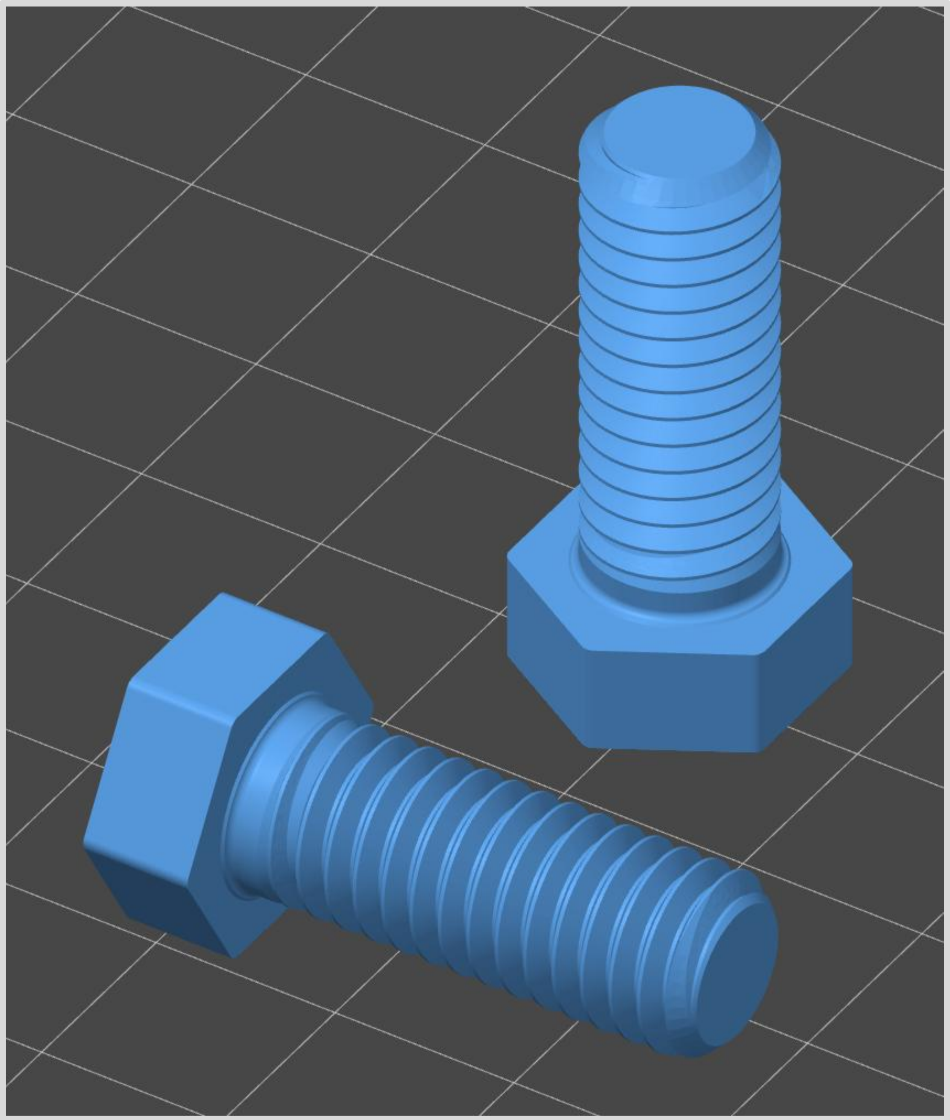
- “Text looks best when indented into a vertical surface. It reduces overhangs and has a better resolution.”

(Tips by Billie Ruben)

Critical Thinking: Screws

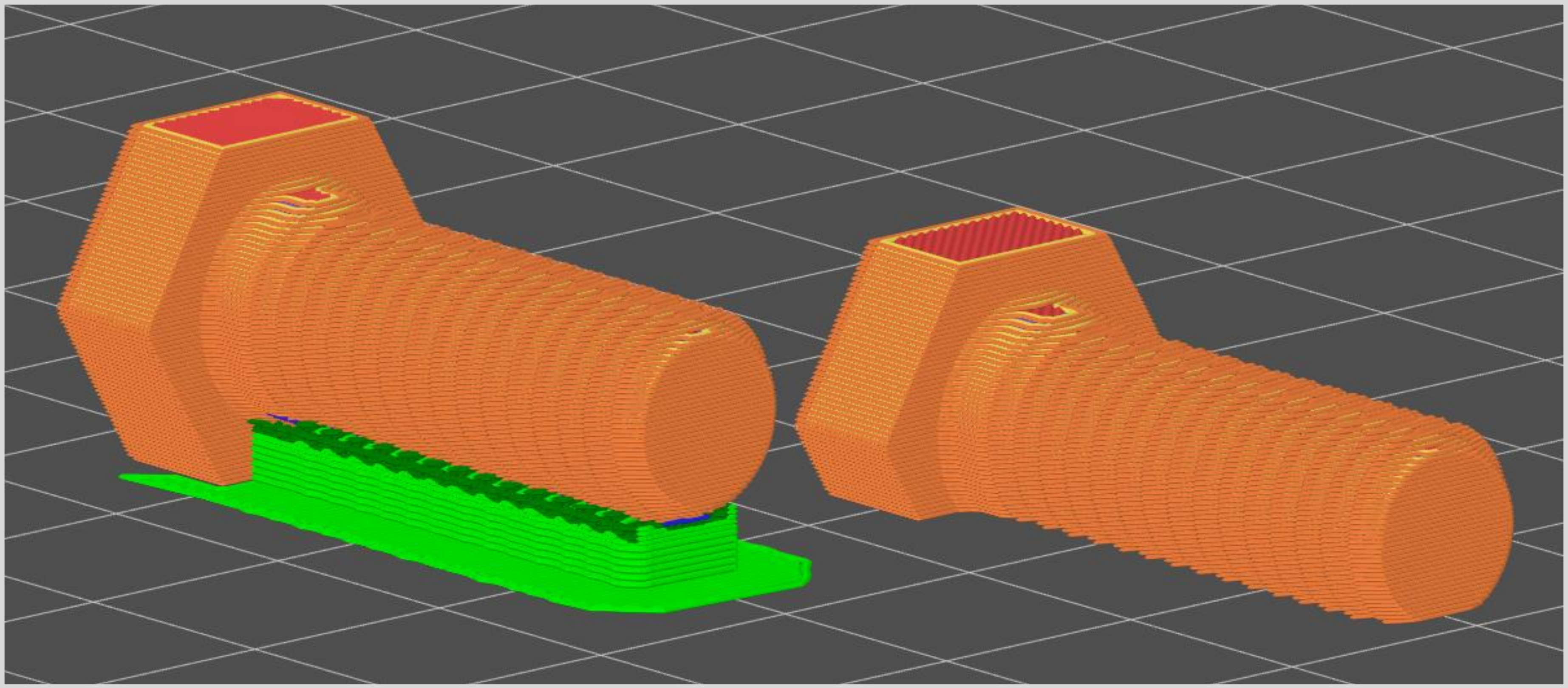
Let's 3D print a plastic screw.

- What differences can you see in strength and printability?
- Compare and contrast print orientations.
- Horizontal.
  - Layer lines maximize strength.
  - Nozzle must be smaller than threads.
  - Bottom threads are floating extrusions. Suggestions?



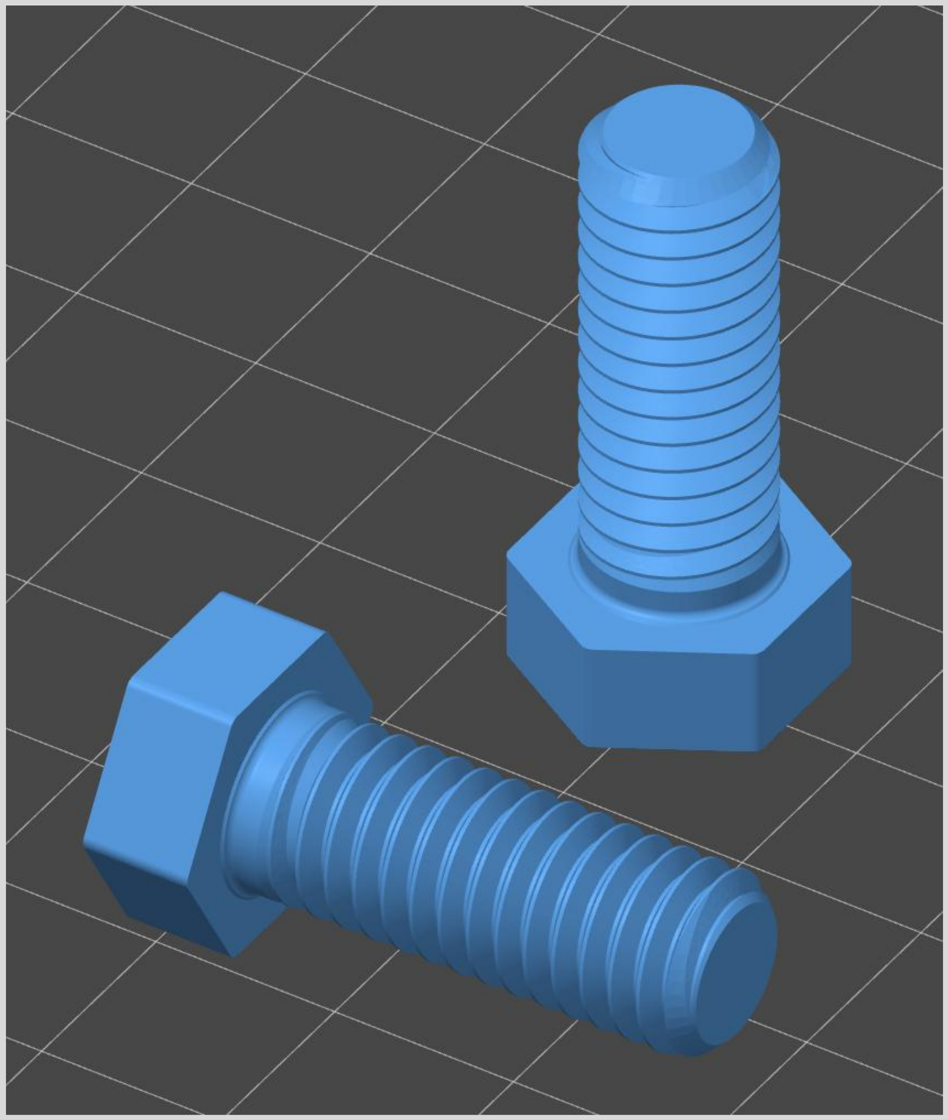
Critical Thinking: Screws

- Supports will probably leave a messy surface.
- Trim the bottom so it can sit flat. Prints clean with lots of bottom surface area.



Critical Thinking: Screws

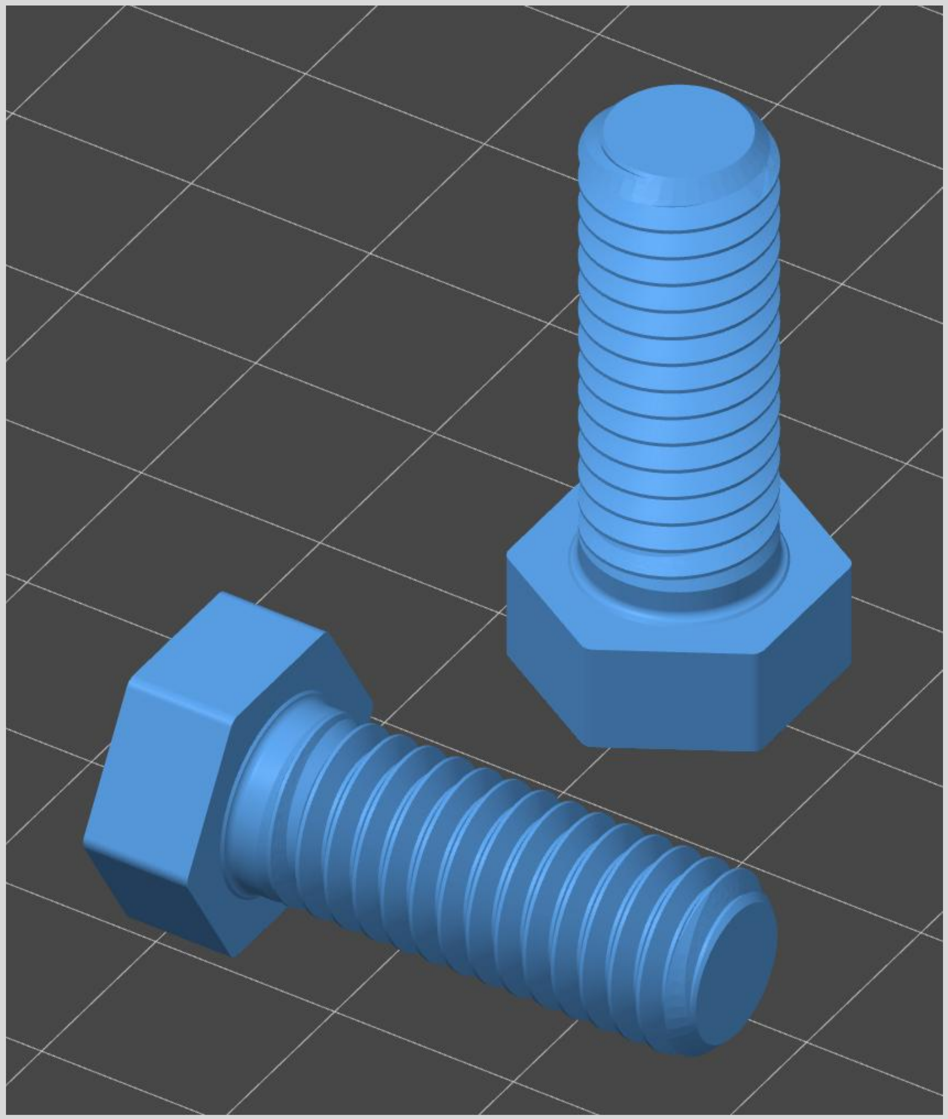
- Vertical
  - Thread size limited by layer height.
  - Steep overhangs for typical threads.
  - Layers lines create a weak part.
  - No supports.
  - No modifications required.



Critical Thinking: Screws

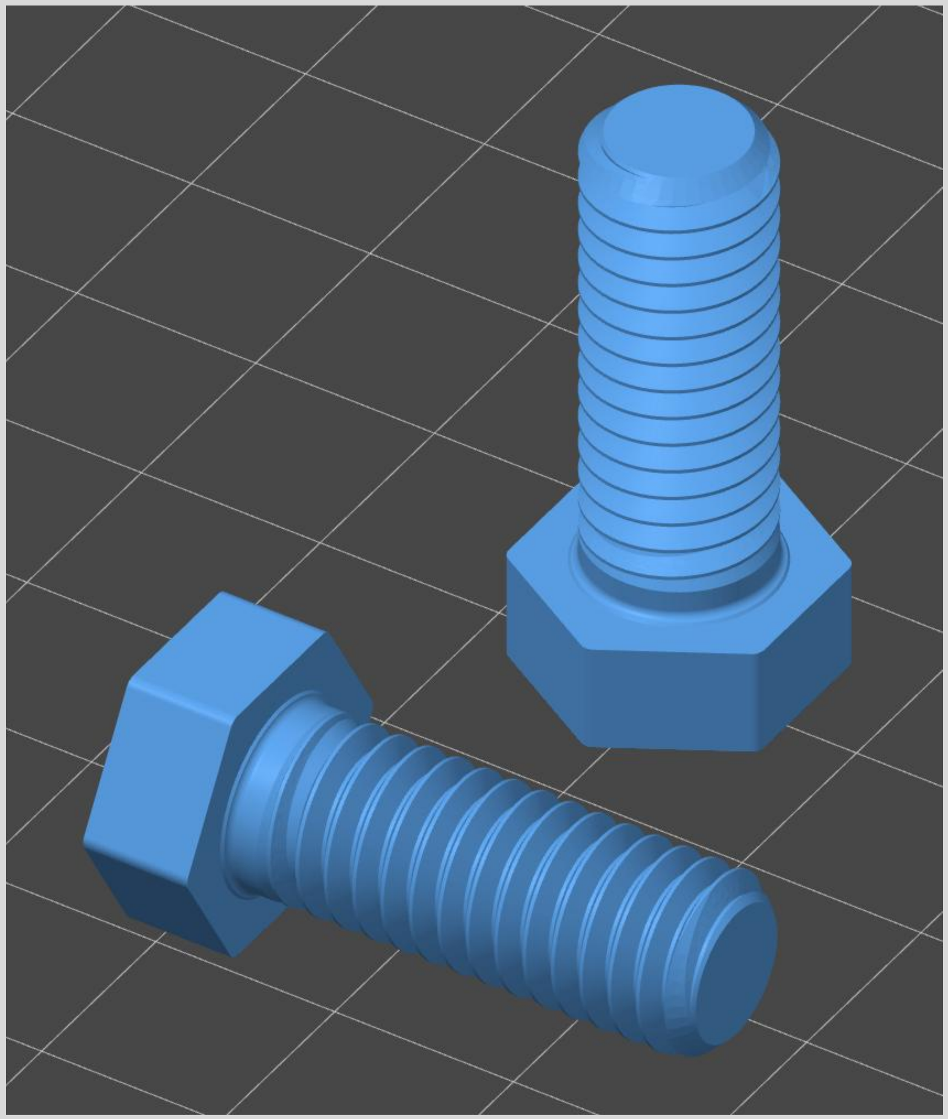
Let's summarize.

- Vertical version.
  - Easy to print.
  - Looks the best.
  - Weakest.
- Horizontal version.
  - Strongest.
  - Printability concerns.



Critical Thinking: Screws

- Compromises like this are common in 3D printing.
- Custom Threads for 3D printing by printables.com/@FlyingGyroscope.
  - Includes CAD lesson.
- Recommended viewing.
  - Strength testing printed bolts by My Tech Fun.
  - And a follow-up video.



Critical Thinking: Snap Buckle

Let’s design a replacement part.

- Should you print a direct copy? Or does it need modifications?
- What 3D printing advantages and disadvantages do you see?
- What print orientation will you use?
  - Compare strength and printability, like screws in the previous example.

(Picture from HomeDepot)

Critical Thinking: Snap Buckle

- Why is the printed buckle designed like that?
- Why did they make modifications?

(Picture from HomeDepot)

(Snap buckle by Squinn)

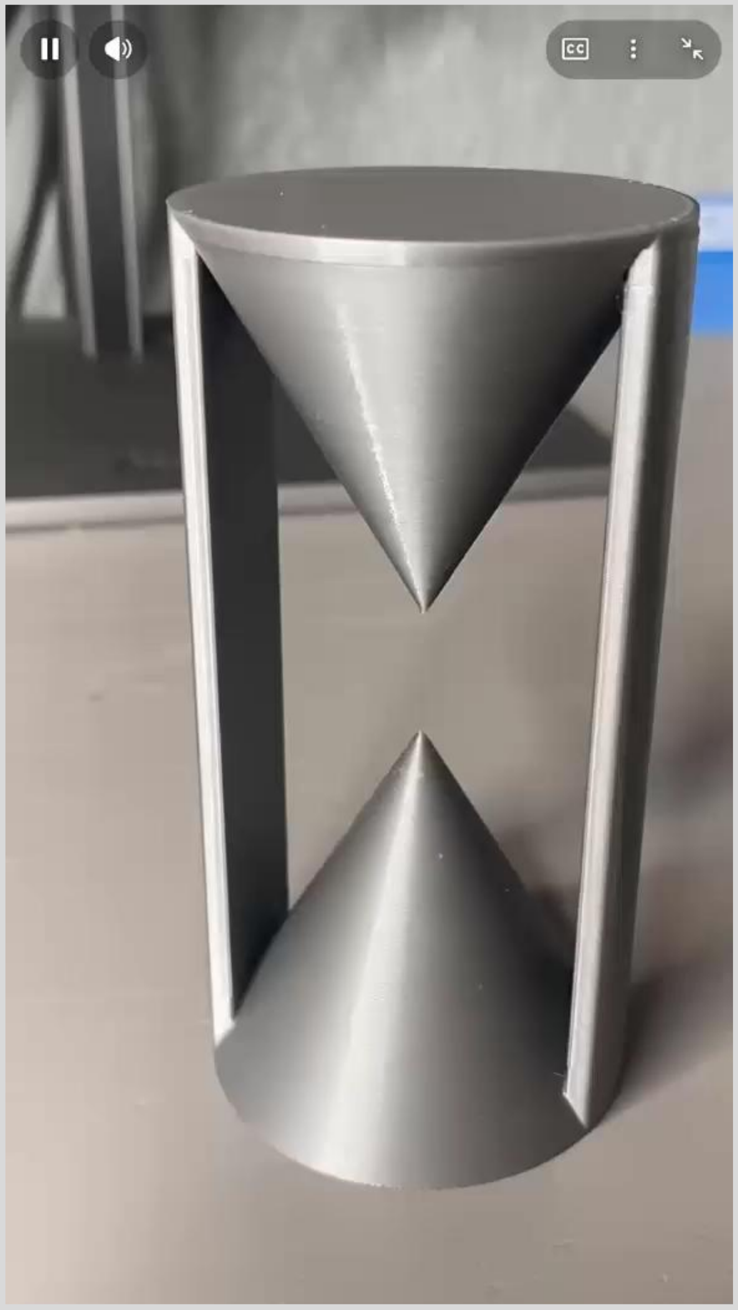
Critical Thinking: Snap Buckle

- Print parts laying down.
  - High strength for intended use.
  - Ideal for the compliant mechanism.
  - Good first layer contact.
- The female socket has been reimagined.
  - Dovetail (trapezoid) traps middle prong.
  - Overhangs instead of a bridge.

(Snap buckle by Squinn)

Creative Solutions

- I love the creativity and innovation of the maker community.
- This was printed without supports. How?
  - Put a printed part inside a printed part.

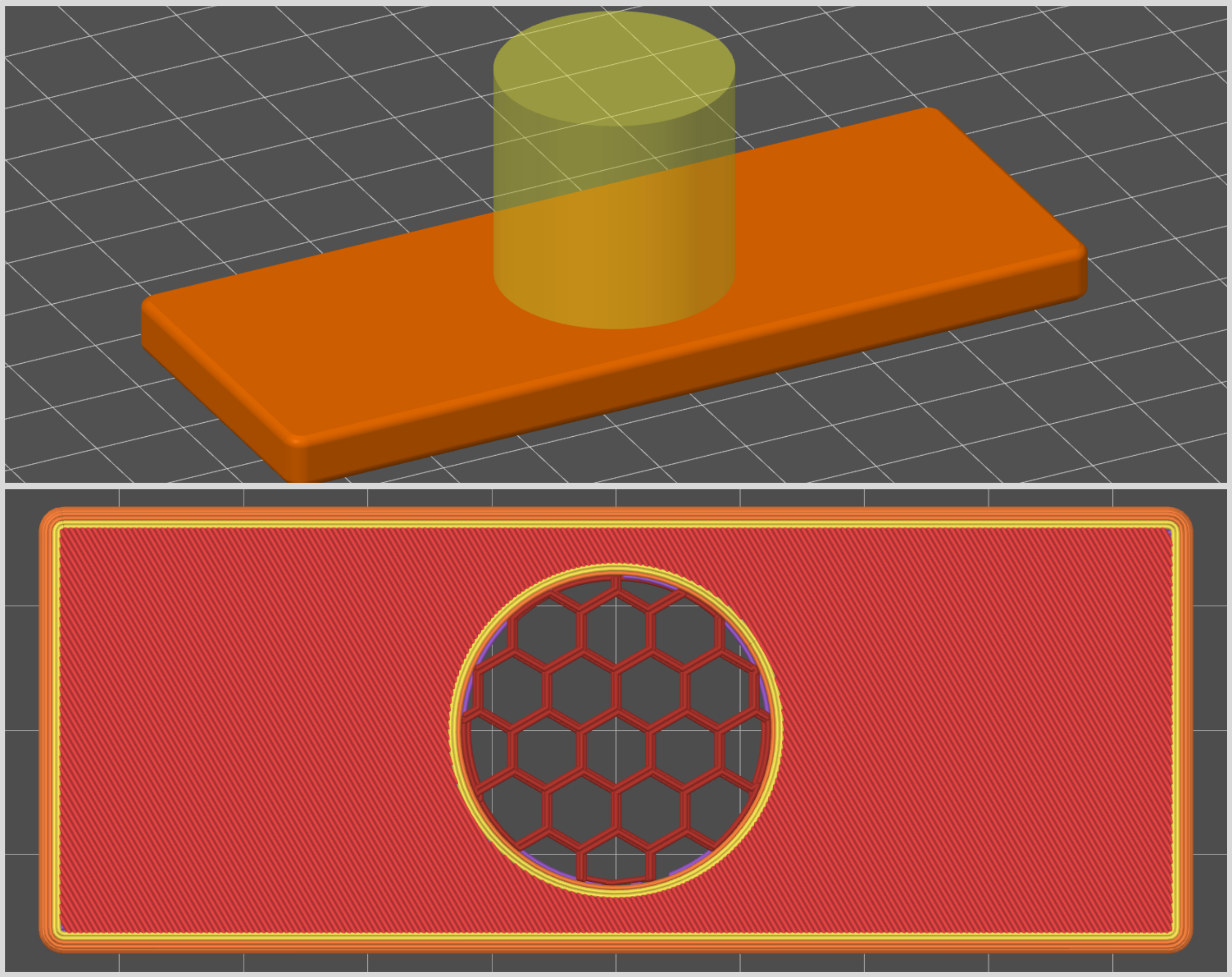


(Animation, YouTube short by christopherhelmke)

Advanced Slicing: Modifiers

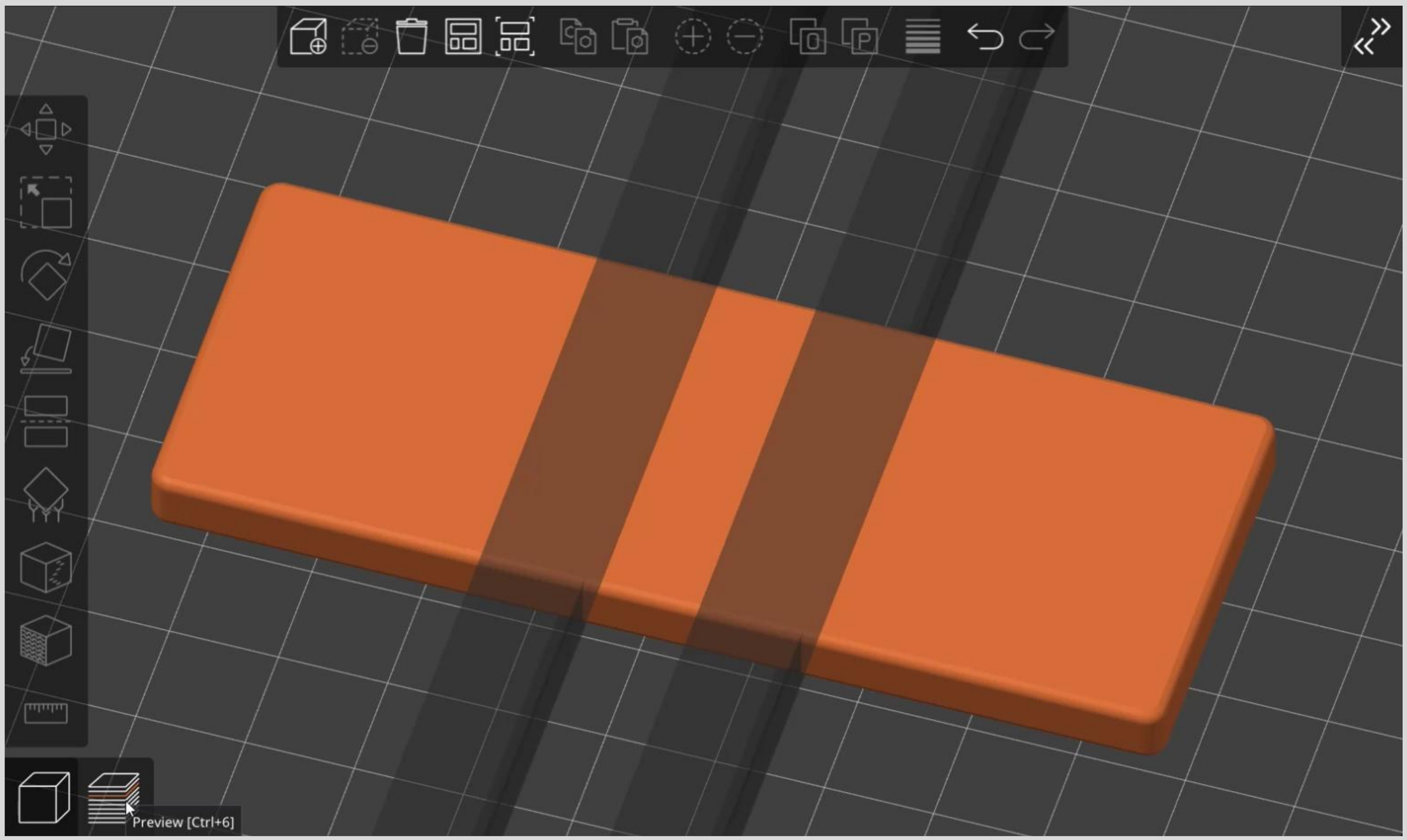
Let's make a mesh grill with PrusaSlicer.

- Add a cylinder part.
- Change it to a modifier.
- Assign zero top and bottom layers.
- Set an infill pattern.
  - Hexagon makes a nice mesh.
- Change infill percent to alter the mesh size.



Advanced Slicing: Negative Volume

- Flexify template. Defines negative space for PIP hinges.
- Start with a flat object.
- Cut away material with templates.



(Animation, FlyingGyroscope GitHub)


THANK YOU
imgflip.com